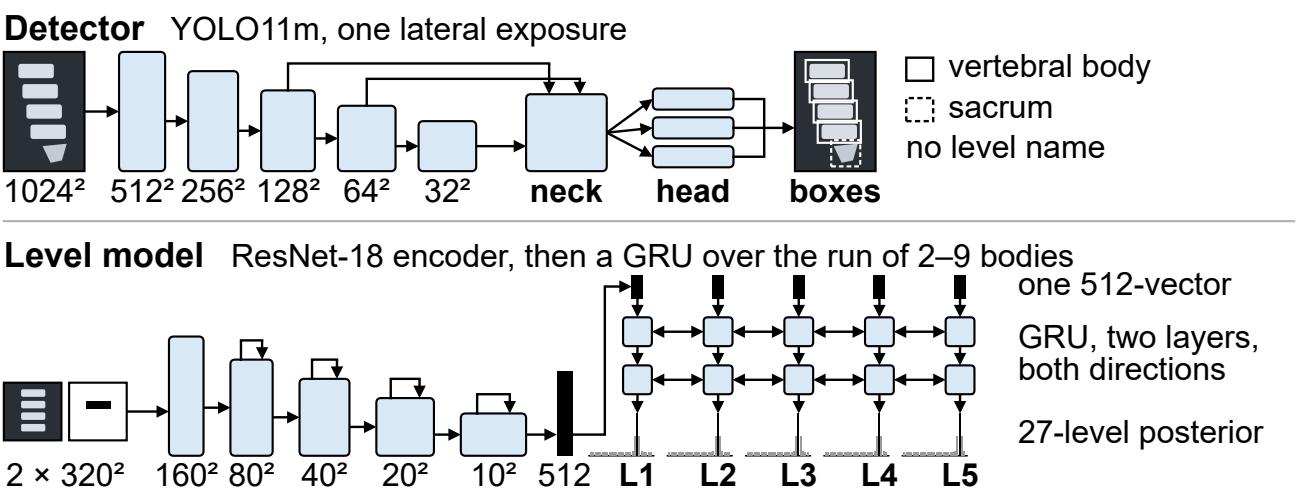
Detector YOLO11m, one lateral exposure
vertebral body
sacrum
no level name
1024²
512²
256²
128²
64²
32²
neck
head
boxes
Level model ResNet-18 encoder, then a GRU over the run of 2–9 bodies
one 512-vector
GRU, two layers, both directions
27-level posterior
2 × 320²
160²
80²
40²
20²
10²
512
L1
L2
L3
L4
L5